We have now seen how to prove various limits using the definition. However, the next example shows that proving limits directly from the definition quickly becomes unwieldy.

Example. Prove that $\lim_{x\to 3} x^2 = 9$.

We have just seen that proving the seemingly simple limit $\lim_{x\to 3} x^2 = 9$ already took a fair amount of ingenuity. Calculating something like

$$\lim_{x\to 2} \frac{5(x-1)^2(x+3)}{23x^5(x+1)^2(x+9)^7} \tag{1}$$

therefore seems hopeless.

The good news is, instead of tackling limits like this one head-on, the following theorem will allow us to build complicated limits out of simpler ones. This will make calculating limits like the one in (1) a breeze.

Theorem. (The Limit Laws) Let $c \in \mathbb{R}$ be a constant and suppose the limits $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist. Then

(1) $\lim_{x\to a}(f(x) \pm g(x)) = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x)$

(2) $\lim_{x\to a}(cf(x)) = c\lim_{x\to a} f(x)$

(3) $\lim_{x\to a}(f(x)g(x)) = \lim_{x\to a} f(x) \lim_{x\to a} g(x)$

(4) $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$, provided that $\lim_{x\to a} g(x) \neq 0$

These laws are also valid

- for one-sided limits.
- for limits at infinity (i.e., when $a = \pm\infty$)
- when one of the two limits is infinite, according to the following rules: if x is any real number, then

$$\begin{aligned}
\infty + x &= \infty \\
-\infty + x &= -\infty \\
x \cdot \infty &= \begin{cases} \infty & \text{if } x > 0, \\ -\infty & \text{if } x < 0. \end{cases} \\
x \cdot -\infty &= \begin{cases} -\infty & \text{if } x > 0, \\ \infty & \text{if } x < 0. \end{cases} \\
\frac{x}{\pm\infty} &= 0.
\end{aligned}$$

There are various cases when the limit on the left-hand side of a given Limit Law may exist, but the formula on the right-hand side cannot be used to calculate the limit. In these cases, the right-hand side takes one of the so-called *indeterminate forms*, which include

$$\frac{0}{0}, \frac{\pm\infty}{\pm\infty}, 0 \cdot \pm\infty, \infty - \infty.$$

None of these expressions can be defined unambiguously. For example, since $x + \infty = \infty$ for every real number x, $\infty - \infty$ has infinitely many solutions. Similarly, if we suppose $\frac{0}{0} = x$, then $0 = x \cdot 0$. But this is true for every real number x.

We now gain some experience with using the Limit Laws calculate limits. First, we revisit the example from the beginning.

Example. $\lim_{x\to 3} x^2$

Example. $\lim_{x\to 5}(2x^2-3x+4)$

Example. $\lim_{x\to -2}\dfrac{x^3+2x^2-1}{5-3x}$